

PROJECT DOCUMENT

BIOMASS ENERGY & AI

Reforestation in Mitsue

Revenue & Payback Summary



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Reforestation in Mitsue

One-Page Funder-Facing Overview

All figures are illustrative early-stage estimates. Phase 1 feasibility studies (due M9, Dec 2026) will replace these with vetted figures anchored to real survey data.

Full detail in [mitsue_implementation_plan.md](#) — "Return on Investment" section.

1. Revenue Streams

Biomass CHP (fuelled by sugi forest thinnings) is the **primary energy source**, providing 24/7 baseload electricity and heat. Solar, battery storage, and EV charging are complementary. The circular local energy economy — where the forest powers the village — underpins all revenue streams.

The plant is an exporter first — size the revenue accordingly. The CHP is sized to the *forest* (~1.15 MWe, ~8,000 MWh/yr), not to any local load. The co-located **data center is a small, stable 24/7 baseload (~18 kW ≈ ~2% of output)** — valuable because it is behind-the-meter (cap-free, worth avoided-retail not export price) and anchors CHP stability, but it absorbs almost nothing. **The bulk of generation goes to the village (via the Kansai T&D grid) and to grid export.** So grid interconnection and Henry's 事前相談 are **central**, and at ~1.15 MW the plant is over the FY2026 <1,000 kW new-FIT cap → most output likely sells via **FIP** (market premium), *not* the 40 円 FIT. A separate ≤1,000 kW line *could* take FIT, TBD. See [../energy-forest/mitsue_fit_grid_check.md](#) (figures provisional, pending primary confirmation).

This ~2% figure is the CHP's own electrons only — it does not cap total DC hosting revenue.

(Reconciled 2026-07-17.) The siting decision in [mitsue_fit_grid_check.md](#) puts compute in **both** places: a small anchor cluster behind-the-meter at the CHP (the ~2% above, valued for stability + avoided-retail), **and** the larger flagship showcase DC at the old school, powered by grid/green retail. So "Data center hosting fees" in the table below is driven by GPU fill-rate across both sites, not by the CHP's export share — see the compute-vs-FIP economics immediately below.

Revenue / Savings Category	Year 1	Year 5	Year 10	Notes
Data center hosting fees	¥0	¥15–30M	¥40–80M	Edge computing / AI specialist workloads

Revenue / Savings Category	Year 1	Year 5	Year 10	Notes
Electricity export (FIP/FIT)	¥0	¥5–15M	¥10–30M	Plant's main energy-revenue stream (village + grid export); > 1,000 kW → likely FIP , a ≤1,000 kW slice may take 40 円 FIT. Magnitude provisional — re-model in Phase 1 once FIP terms + village offtake are set
EV charging fees	¥0	¥1–3M	¥3–8M	Growing as EV fleet expands
Carbon credits (J-Credit)	¥0	¥1–3M	¥3–10M	From native forest restoration
Forestry products (timber, lumber)	¥0	¥3–8M	¥10–20M	Beyond fuel-residue use
Education / consulting ("playbook")	¥0	¥1–3M	¥3–8M	Replication licensing and training
Total annual revenue (illustrative)	¥0	¥28–67M	¥74–166M	

Why "compute" ranks above "electricity export": 1 kWh is worth ~6–20× more as AI compute than as exported power. (Added 2026-07-17.)

Use of 1 kWh	Revenue	Basis
Sell to grid (FIP)	~¥30–40	未利用材 biomass FIP effective (基準価格 ~32–40円 + JEPX premium)
Village displacement	~¥20–30	avoided retail import
Feed an A100 GPU (GPUSOROBAN retail)	~¥360–650	¥361/hr ÷ ~0.55–1 kW/node
Feed 8×H200 server (retail → 50%-off enterprise)	~¥390 → ~¥190	¥2.78M → ¥1.39M /mo ÷ 720h ÷ ~10 kW

Consequence: the CHP's best "customer" is an **on-site GPU DC, not the grid**. Grid export (FIP) is the **overflow valve** for power we can't yet convert to compute — a revenue *floor*. Size the DC as large as we can (a) fill with paying GPU demand and (b) fit+cool at 神末. The scarce factors flip from power to **GPU capex + fill-rate**: a rented A100 earns ~¥360/kWh, an idle one ¥0 while ~¥1.3M capex depreciates → **utilization, not power, is the constraint** (→ HIGHRESO partnership / offtake pipeline). Sources: GPUSOROBAN pricing <https://soroban.highreso.jp/>; FIP/未利用材 per [../energy-forest/mitsue_fit_grid_check.md](https://energy-forest/mitsue_fit_grid_check.md).

1a. Net-of-capex reality check — compute only beats FIP above a breakeven fill-rate

(Added 2026-07-17.) The ~6–20× figure above is **gross revenue per kWh** — it doesn't net out GPU capex/depreciation/opex, which FIP export doesn't carry (the CHP is already sunk capex either way). On a **net-**

profit basis the comparison is capex- and utilization-sensitive, so it belongs in the doc as a breakeven condition, not a single multiple.

Assumptions (sourced, not invented):

Input	Value	Source
8×H200 server capex	~¥53M (range ¥46–60M)	\$320–420K typical \$370K, ×¥143/\$ — Mercatus H200 server price
Power draw per server	~10 kW	Existing GPUSOROBAN-based calc, this doc §1
Hosting revenue, list	¥2.53M/mo (¥30.36M/yr @ 100% util)	HIGHRESO GPUSOROBAN H200 pricing
Hosting revenue, long-contract discount	up to –39% → ¥1.54M/mo (¥18.52M/yr @ 100% util)	Same source
Opex	~20% of realized revenue	Planning assumption (typical colo/cloud opex ratio) — not vendor-quoted
Depreciation life	4 yr (small-operator planning) vs 6 yr (hyperscaler accounting standard)	CNBC — AI GPU depreciation debate ; AWS/Google/Microsoft moved 3–4yr → 6yr in 2023–24
FIP-equivalent net (same ~87,600 kWh/yr, no meaningful capex)	~¥3.07M/yr	¥30–40/kWh mid ¥35, this doc §1

Breakeven utilization — the fill-rate a server needs to beat simply exporting its power via FIP:

Price tier	Depreciation life	Utilization needed to beat FIP
List price	4 yr	~67%
List price	6 yr	~49%
Long-contract discount (–39%)	4 yr	>100% — can never beat FIP at this price/life
Long-contract discount (–39%)	6 yr	~80%

The finding: netting real GPU capex reverses the naive "compute always wins" reading of the 6–20× gross figure above. A GPU server only beats simply exporting its power once it clears roughly **45–70% fill-rate at list pricing** — and essentially **never** at the deep long-term discount unless depreciation stretches to the 6-year hyperscaler standard. This does not contradict §1's per-kWh economics (compute is still the higher-value use of a kWh) — it says the *server*, as a ¥46–60M capital deployment, needs a real utilization floor to convert that per-kWh advantage into net profit. Grid-export FIP requires no such floor because the CHP capex is already committed regardless.

Open question — not yet answered: what fill-rate can the HIGHRESO offtake pipeline actually underwrite? That number (not a guess) picks which row of the table above is the honest one to plan and pitch against. Until confirmed, treat the anchor-compute buildout size as **fill-rate-gated**, not power-gated — matching `../energy-forest/mitsue_fit_grid_check.md`'s "utilization, not power, is the constraint," now quantified.

2. Cost Displacement

The project *replaces or avoids* significant existing costs:

- **Energy imports:** Mitsue residents and businesses currently import ~100% of electricity. Local generation displaces approximately **¥40–60M/yr** of energy spending leaving the village economy.
- **Data-center power (self-consumption):** the CHP supplies the co-located data center's electricity behind-the-meter, **avoiding retail grid purchase** — the foundation of the "cheap green compute" thesis. It is a *small* load (~18 kW ≈ ~2% of plant output), so the value is per-kWh (avoided retail > export) and CHP-stability, not volume. Because it is *avoided cost*, not grid sales, it is unaffected by the FIT capacity cap.
- **Former Sugano school maintenance burden:** The Koryukan costs the village an estimated **¥3–8M/yr** in basic upkeep with no return. Active reuse as a data center hub converts a stranded asset into a productive facility.
- **Forest management liability:** Untended sugi plantations impose ecological costs (pollen, biodiversity loss) and physical risks (landslide, fire). Active management converts this liability into feedstock and timber revenue.

3. Payback Framework

Metric	Value
Total capital deployed (Phases 1–3) by Year 5	¥200–290M
Annual revenue by Year 5	¥28–67M
Operating costs by Year 5 (est. 60% of revenue)	¥18–35M
Net annual surplus by Year 5	¥10–32M
Approximate capital payback period	10–18 years (depending on grant vs. loan vs. revenue-financing mix)
Break-even / net-surplus year (base case)	~Year 5

Once net surplus is achieved, reinvestment priority: (1) forestry scale-up → (2) data-center expansion → (3) replication support for other villages.

4. Honest Caveat

These are **framework figures, not a forecast**. They assume successful execution of all three project elements (biomass CHP energy, data center, forestry) at target scale. Phase 1 feasibility studies will sharpen these numbers significantly. Real-world performance — particularly biomass CHP sizing, data center occupancy, and FIT/FIP tariff outcomes — will determine actual ROI.

Sources

These are the project's **own illustrative early-stage estimates**, consolidated from the "Return on Investment" section of [mitsue_implementation_plan.md](#) — not externally audited figures. The underlying load, PUE, and carbon assumptions and their references (HIGHRESO PUE target, J-Credit methodology, Forestry Agency sequestration data, edge-data-center power benchmarks) are documented in that plan's Phase 1 "Sources & assumptions" block.

The categories above will be validated against METI FIT/FIP tariff schedules, J-Credit pricing, and biomass/timber market data **during** the Phase 1 feasibility studies (M9, December 2026), which will replace these framework figures with vetted estimates.

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